
```
% Julie Hum
% ECE210A Assignment 4

clc;

%1.
dotprod = @(value1, value2) value1' * value2;

%2.
function result = is_orthonormal(twoDarray, func_handle)
    result = false;

    for i = 1:size(twoDarray, 2)

        for j = i+1:size(twoDarray, 2)
            if abs(func_handle(twoDarray(:, i), twoDarray(:, j))) > 1000 *
eps
                return
            end
        end

        if abs(func_handle(twoDarray(:, i), twoDarray(:, i)) - 1) > 1000 *
eps
            return
        end

        result = true;
    end

%3.
function result = gram_schmidt(TDA, func_handle)
    if is_orthonormal(TDA, func_handle)
        result = TDA;
        return
    end

    for i = 1:size(TDA, 2)
        v = TDA(:, i);

        for j = 1:i-1
            v = v - func_handle(TDA(:, j), TDA(:, i)) * TDA(:, j);
        end

        TDA(:, i) = v / sqrt(func_handle(v, v));
    end
    result = TDA;
end

%4.
M = randi(15, 4, 4) + 1j * randi(15, 4, 4);
```

```
x = gram_schmidt(M, dotprod)
is_orthonormal(x, dotprod)
```

```
x =
```

```
    0.2741 + 0.4569i    0.3289 - 0.4153i   -0.2967 - 0.2585i   -0.1969 + 0.4918i
    0.4264 + 0.3655i    0.1499 - 0.2812i    0.5639 + 0.3214i    0.0095 - 0.4021i
    0.4264 + 0.1827i   -0.1205 + 0.7219i    0.1671 + 0.1628i   -0.0960 + 0.4307i
    0.3046 + 0.3046i   -0.0839 + 0.2740i   -0.3693 - 0.4827i    0.3044 - 0.5198i
```

```
ans =
```

```
logical
```

```
1
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